

1720

MTL/NDT/2026

**REPORT ON AN IN-SITU INTEGRITY TEST
(NON-DESTRUCTIVE) OF COMPRESSIVE
STRENGTH OF STRUCTURAL ELEMENTS**

BY

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ON

Botanical Garden Road, Ebute Metta

FOR

-

Botanical Garden Road, Ebute Metta, Lagos, Lagos Mainland, Lagos

DATE OF TEST: NOT RECORDED

DATE OF REPORT: 04/09/2026

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1.0 INTRODUCTION

This report presents the findings of a non-destructive (ultrasonic pulse velocity) integrity investigation carried out on the project "Botanical Garden Road, Ebute Metta". The investigation was conducted using the Portable Ultrasonic Non-Destructive Digital Indicating Tester (PUNDIT) in accordance with BS 1881-203:1999 and ASTM C597.

Project Reference	NXC-GOV-2026-522F
Project Number	-
Site Address	Botanical Garden Road, Ebute Metta, Lagos
Local Government Area	Lagos Mainland
State	Lagos
Structural System	-
Number of Floors	-
Development Category	-
Structural Elements Tested	1
Ultrasonic Tests Recorded	1

2.0 PURPOSE OF INVESTIGATION

The purpose of the investigation is to:

- Determine the present status of the tested structural elements from the outcome of the ultrasonic non-destructive tests, with respect to BS EN 12504-4:2021 and ASTM C597-09.
- Determine the ongoing concrete quality of the tested structural elements with respect to BS 8110: Part 1 1997, using the laboratory velocity-strength calibration stated in Section 3.0.
- Provide engineering advice based on the recorded ultrasonic indications and the visual observations made during the field work.

Recorded tests comprise 1 pulse velocity test(s), 1 crack depth test(s) and 0 surface quality observation(s).

3.0 LITERATURE REVIEW

The ultrasonic pulse velocity method is a well-established non-destructive technique for assessing the quality and homogeneity of concrete. A pulse of longitudinal vibrations is produced by an electro-acoustical transducer held in contact with one face of the concrete; a similar transducer on the opposite face receives the pulse, and the time taken for the pulse to traverse the concrete is measured electronically. The pulse velocity is then the path length divided by the transit time.

Velocity is influenced by the elastic modulus and density of the concrete, and hence by its strength and condition. Following Whitehurst's classification (adopted in BS 1881-203:1999 and ASTM C597), velocities of 4.5 km/s and above indicate excellent quality; 3.75 - 4.5 km/s good; 3.0 - 3.75

km/s questionable; 2.0 - 3.0 km/s poor; and below 2.0 km/s very poor.

Estimated compressive strength (E.C.S) values in this report are derived from the laboratory's fixed ultrasonic pulse-velocity calibration curve $f_{cu} = 8.96 \times V - 7.97$ (f_{cu} in N/mm², V in km/s), established by least-squares regression over the laboratory's reference control pairs (2.9, 19), (3.9, 25), (4.0, 27), (4.2, 29) and (4.4, 34) and valid over the range 2.0 - 5.0 km/s. BS 1881-203:1999 notes that no unique velocity-strength relationship exists for all concretes; the curve above is the documented calibration applied by this laboratory for Nigerian site concrete and is applied uniformly to every result in Section 5.0. Velocities outside the calibrated range are reported without an E.C.S estimate rather than extrapolated.

4.0 LOCATION OF SITE, WEATHER CONDITION AND FIELD WORK / EQUIPMENT STATUS CHECK

Site Address	Botanical Garden Road, Ebute Metta, Lagos
Ward / Area	-
Local Government Area	Lagos Mainland
State	Lagos
Weather Condition	NOT RECORDED (no weather observation model exists on the platform - the platform does not fabricate weather data)

No field device was recorded against these tests.

4.1 VISUAL TEST

- Floor:200THK RC SLAB:780904: [MANUAL_FIELD_ENTRY - Station UPV-FLD-2026-002] tacky floor
- M_Footing-Rectangular:900 x 900 x 200mm:803711: [MANUAL_FIELD_ENTRY - Station fghjkjhg] helllo

4.2 METHODOLOGY

Non-destructive concrete strength determination was carried out using the Portable Ultrasonic Non-Destructive Digital Indicating Tester (PUNDIT). The pulse velocity V is computed as the path length L divided by the transit time t ($V = L / t$, with L in mm and t in microseconds, so V is in km/s). Concrete quality is graded against the BS 1881-203 / ASTM C597 velocity bands stated in Section 3.0.

Tests were carried out with 54 kHz transducers, as recorded against each test.

Crack depth was determined by the time-difference method of BS 1881-203: $d = (L / 2) \times \sqrt{(t_{cracked} / t_{uncracked})^2 - 1}$, where L is the path length between transducers straddling the crack.

Estimated compressive strength conversion: E.C.S: $f_{cu} = 8.96 \times V - 7.97$ (f_{cu} in N/mm², V in

km/s; valid 2.0 - 5.0 km/s). The full calibration statement is given in Section 3.0.

4.3 REINFORCING BAR (REBAR) ASSESSMENT

No dedicated reinforcing bar cover measurements (e.g. Profoscope scans) were recorded during this investigation. Observations are limited to the ultrasonic indications reported in Sections 4.1 and 5.0; measured crack depths that may affect concrete cover are reported in the crack depth table of Section 5.0.

4.4 EQUIPMENT / REBAR ASSESSMENT TABLE

No field device was recorded against these tests.

REBAR ASSESSMENT TABLE (Profoscope LSM/TL/Prof/026/003)

No Rebar scanning data recorded for this project.

NOTE: This assessment is non-destructive and does not replace construction reinforcement design drawings.

Crack depth measurements are tabulated in Section 5.0 (time-difference method).

5.0 ANALYSIS OF TEST RESULTS

1 test(s) have not been run through the platform analysis endpoint; velocities shown are computed directly from the recorded measurements using the same deterministic BS 1881-203 relations.

Element average E.C.S is computed from the element mean pulse velocity through the Section 3.0 calibration curve. Velocities outside the calibrated 2.0 - 5.0 km/s range are reported without an E.C.S estimate rather than extrapolated.

STRUCTURAL ELEMENT: M_Footing-Rectangular:900 x 900 x 200mm:803711

S/N	TEST REF	PATH (MM)	TRANSIT TIME (US)	PULSE VELOCITY (KM/S)	E.C.S (N/MM2)
1	PND-2026-4FAC71	300.0	56.0	5.36	-

AVERAGE PULSE VELOCITY: 5.36 KM/S AVERAGE E.C.S: - N/MM2 REMARK: EXCELLENT

5.1 CRACK DEPTH MEASUREMENTS (TIME-DIFFERENCE METHOD)

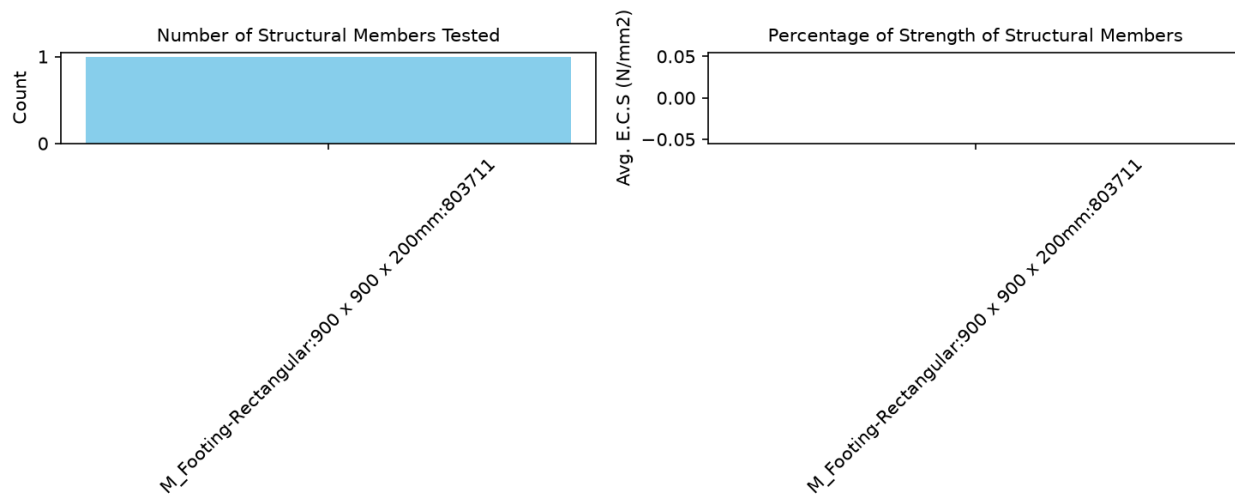
ELEMENT	PATH (MM)	T CRACKED (US)	T UNCRACKED (US)	CRACK DEPTH (MM)	REMARK
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ELEMENT	PATH (MM)	T CRACKED (US)	T UNCRACKED (US)	CRACK DEPTH (MM)	REMARK
Floor:200THK RC SLAB:780904	300.0	70.0	62.5	75.7	Depth exceeds 25 mm - structural review required

6.0 RECOMMENDATIONS

- [HIGH] Map crack extent and monitor; engage structural engineer for crack-depth exceedance.
- [URGENT] Immediate structural engineering review - pulse velocity indicates compromised concrete.

BAR CHART SHOWING SUMMARY OF TEST RESULTS (BOTANICAL GARDEN ROAD, EBUTE METTA)



Legend: GFC: Ground Floor Column | FFC: First Floor Column | FFB: First Floor Beam | FFS: First Floor Slab

7.0 CONCLUSION

A total of 1 ultrasonic pulse velocity test(s) were carried out on 1 structural element(s). The overall mean pulse velocity was 5.36 km/s (minimum 5.36 km/s, maximum 5.36 km/s).

Elements graded by BS 1881-203 velocity bands: 1 excellent.

All tested elements fell within or above the "good" velocity band at the time of test.

8.0 REPORT INTEGRITY AND SIGN-OFF

This report was generated from verified platform records. Every figure is computed from live test records; the content digest below is a SHA-256 hash of the underlying test identifiers and results and can be used to detect any later alteration.

Report No	1720 / MTL/NDT/2026
Tested by	Abdullateef, Abdullateef Ojugbele
Approved by	nexucon nexucon _____
Content digest	50fff9f4651758e9e28ec1d0ea5de6544deae738089479955d87125a6ca62ed9
Date	2026-09-04 15:12

APPENDIX PHOTOGRAPHS



Photo 1: 1786628996237056000.jpg (Floor:200THK RC SLAB:780904)



Photo 2: 1786628911732759040.jpg (Floor:200THK RC SLAB:780904)



Photo 3: 1786628988736631808.jpg (Floor:200THK RC SLAB:780904)



Photo 4: 1786628977736079872.jpg (Floor:200THK RC SLAB:780904)



Photo 5: 1786628993736943872.jpg (M_Footing-Rectangular:900 x 900 x 200mm:803711)



Photo 6: 1786628998237142784.jpg (M_Footing-Rectangular:900 x 900 x 200mm:803711)



Photo 7: 1786628996237056000.jpg (M_Footing-Rectangular:900 x 900 x 200mm:803711)



Photo 8: 1786628977736079872.jpg (M_Footing-Rectangular:900 x 900 x 200mm:803711)